

Beyond Word-Level Sentiment

Structural improvements to NLP alpha-signal generation

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The dominant production text-to-signal pipelines in systematic equity investing — supervised sentiment in the style of SESTM (Ke, Kelly & Xiu), Loughran–McDonald dictionaries, and commercial sentiment feeds — share three structural assumptions that can limit the marginal quality of the resulting signals. This proposal relaxes all three and reframes the output from a stock *ranking* into calibrated signal representation ($\hat{\mu}_i, \hat{\sigma}_i$) that downstream teams can consume directly. Each direction was motivated by a concrete limitation the author observed while replicating the corresponding published method.

The three assumptions we target

- (1) **Firm-invariant language** — the return implication of a term is assumed to be identical across firms.
- (2) **Time-invariant language** — the return implication of language is assumed to remain stable across market regimes.
- (3) **Rank-only output** — the signal provides relative rankings but not calibrated measures of expected return or confidence.

Four research directions

- Firm-conditioned sentiment, OURS** — move from a market-wide map to firm-conditioned sentiment at the topic/event-cluster level, via a low-rank firm $\times$ cluster model with characteristic-based shrinkage (clusters, not raw words, to keep estimation feasible). The return implication of a term may depend on firm characteristics, industry context, and balance-sheet conditions. For example, a term such as “restructuring” may signal operational improvement for a distressed firm but carry negative return implications for a mature firm undertaking defensive cost cutting. A single market-wide sentiment score implicitly averages across these heterogeneous interpretations.
- Regime-conditioned language, OURS** — let sentiment depend on an ex-ante macro state z_t (volatility, rates, business cycle conditions) through an additive cluster-level term. For example, “cost reduction” may be rewarded during periods of economic stress but receive little incremental attention during strong expansions. A static sentiment model averages across these changing economic environments.
- From ranking to calibration, OURS** — calibrate scores into expected-return estimates with an associated confidence measure, ($\hat{\mu}_i, \hat{\sigma}_i$), using robust, heavy-tail-aware objectives (Huber / quantile / Student- t , with isotonic calibration). Rather than producing rankings alone, the signal provides calibrated estimates of expected return and confidence. Portfolio sizing and optimisation remain the responsibility of downstream teams.
- Production alpha pipeline, OURS** — develop a point-in-time workflow from multi-source text (newswire, earnings-call transcripts, regulatory filings, and alternative text data) to a versioned signal-delivery contract, with leakage controls, signal-decay monitoring, and retraining diagnostics embedded as first-class production components.

Production-maturity extensions: event-specific alpha-decay *half-lives* (each signal is accompanied by an estimated information horizon, not merely a directional score); *blind/anonymized-input* validation (textual content vs company memorisation); and *concept-drift* monitoring with scheduled/triggered retraining.

Honest framing of the evidence

Published NLP strategies have reported strong performance — for example, Ke, Kelly & Xiu (2019) report a daily-news long–short Sharpe ratio above 4, **PRIOR-LIT**. However, such results are typically gross, high-turnover, and concentrated in smaller stocks. More recent work has further shown that LLM-based alpha can be materially overstated under contaminated or non-point-in-time evaluation, **PRIOR-LIT**. Accordingly, the focus of this proposal is not larger models, but more robust signal construction through firm-conditioned and regime-conditioned language representations.

How this proposal is framed

This is an external research agenda, judged on signal quality, tradability, and production realism rather than novelty for its own sake. The comparative advantage of this agenda is not familiarity with the literature but hands-on **replication** of these pipelines, **diagnosis** of where they strain in practice, and concrete proposals to **improve** them — expressed as point-in-time signals with explicit confidence that downstream sizing, risk, and execution teams can consume.

Companion documents: **Research_Proposal.pdf** (2-page pitch) and **Appendix.pdf** (methodology, derivations, replication notes, current-literature scan). Performance figures cited anywhere in this package are reference results from prior published literature, gross unless stated, and are not a track record of the author. This is the author's independent personal work, not affiliated with or endorsed by any institution or employer.